Contribution Summary

Name: Lipsa Sahoo

College: Easwari Engineering College

Batch: 2026

Role & Initial Responsibilities

During my onboarding, I was assigned the role of a DevOps engineer. However, based on project requirements, I initially worked on QA activities to understand the testing side of the software development lifecycle.

Learning & QA Exploration

In the initial phase, I explored key QA tools and practices. I worked with Playwright for automation testing, where I learned to design end-to-end test cases using best practices like POM and saved authentication. I also gained exposure to load testing, understanding how to evaluate system performance under different workloads.

Hands-on Testing Experience

After building a strong foundation, I applied my knowledge by testing two newly developed features:

- RTM (Requirement Traceability Matrix)
- Audit Tracker

I was responsible for validating functionalities, identifying defects, and ensuring that the features met business and technical requirements. This gave me real-world exposure to QA workflows and improved my debugging and analytical skills.

DevOps Learning & Implementation

As I progressed, I transitioned into core DevOps concepts and tools:

- I learned AWS VPC (Virtual Private Cloud) fundamentals, including networking concepts like subnets, routing, and security groups, which helped me understand how secure and isolated cloud environments are designed.

- I explored Docker, where I understood containerization—how applications and their dependencies can be packaged into containers to ensure consistency across development, testing, and production environments.
- I worked with AWS CodePipeline, learning how CI/CD pipelines automate the build, testing, and deployment processes.

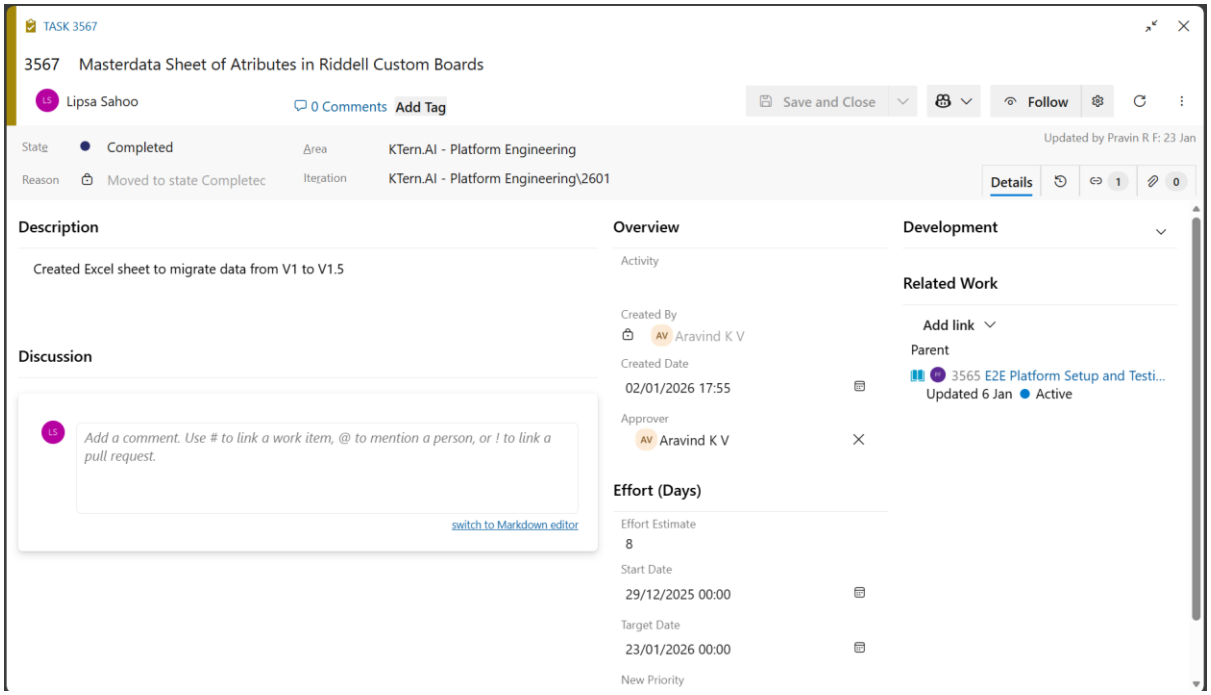
Using these concepts, I contributed to setting up a CI/CD pipeline for the Audit Tracker for Quality, integrating automated testing into the pipeline to ensure code quality and reliability before deployment.

Conclusion

Overall, this journey helped me bridge both QA and DevOps domains. I gained hands-on experience in automation testing, cloud networking (VPC), containerization (Docker), and CI/CD pipelines. This holistic exposure enabled me to understand how modern applications are tested, deployed, and maintained in real-world environments.

Task 1:

https://dev.azure.com/KaariOpEx/KTern.AI%20-%20Platform%20Engineering/_sprints/taskboard/KTern.AI%20-%20Platform%20Engineering%20Team/KTern.AI%20-%20Platform%20Engineering/2601?workitem=3567

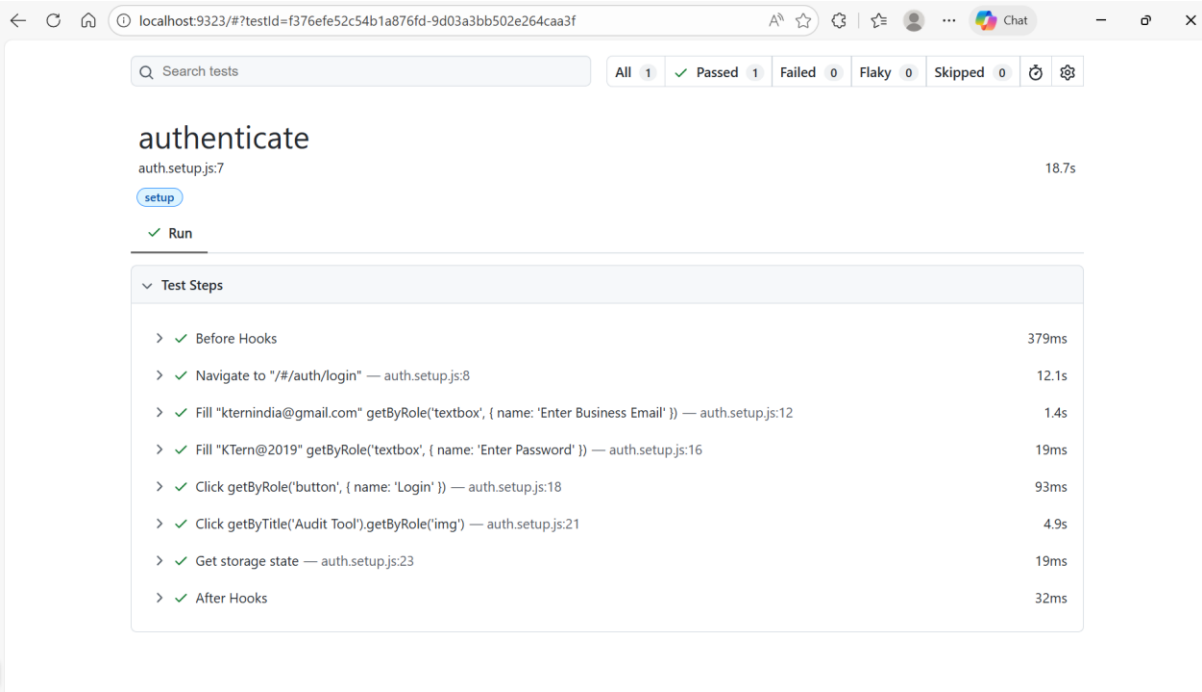


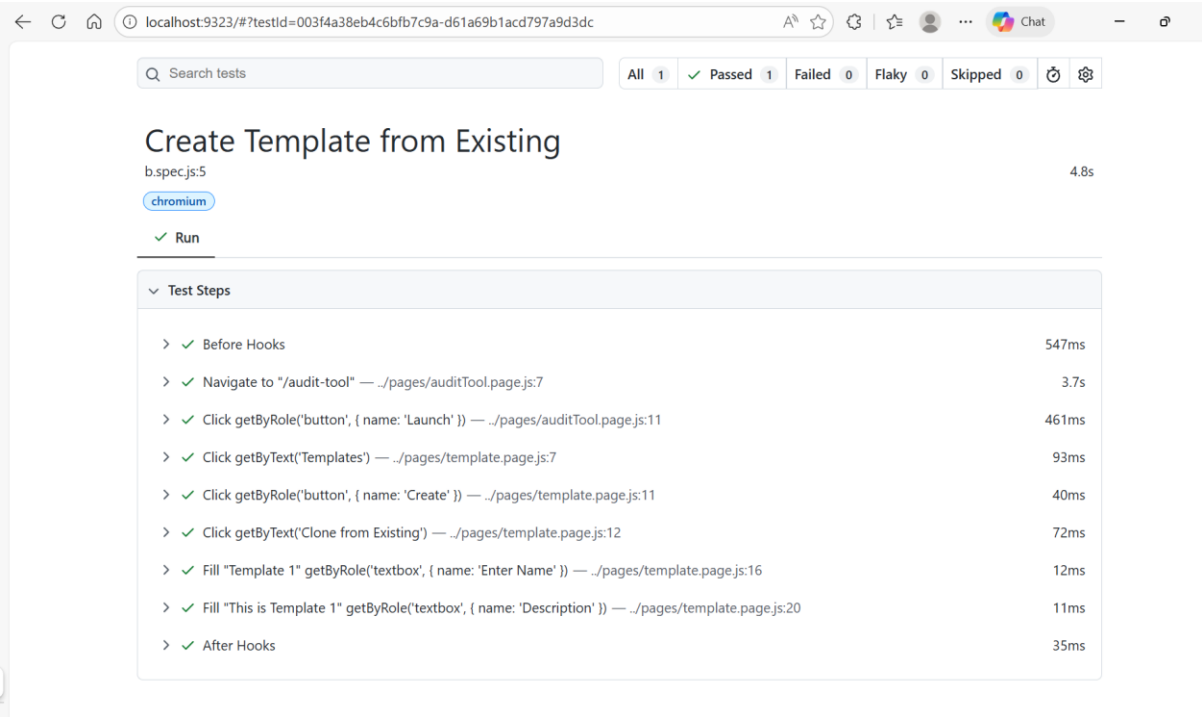
In this the requirement was collecting all the attributes used in Custom boards that will be used to create entity types for Entity.

Task 2:

Audit Tracker Testing

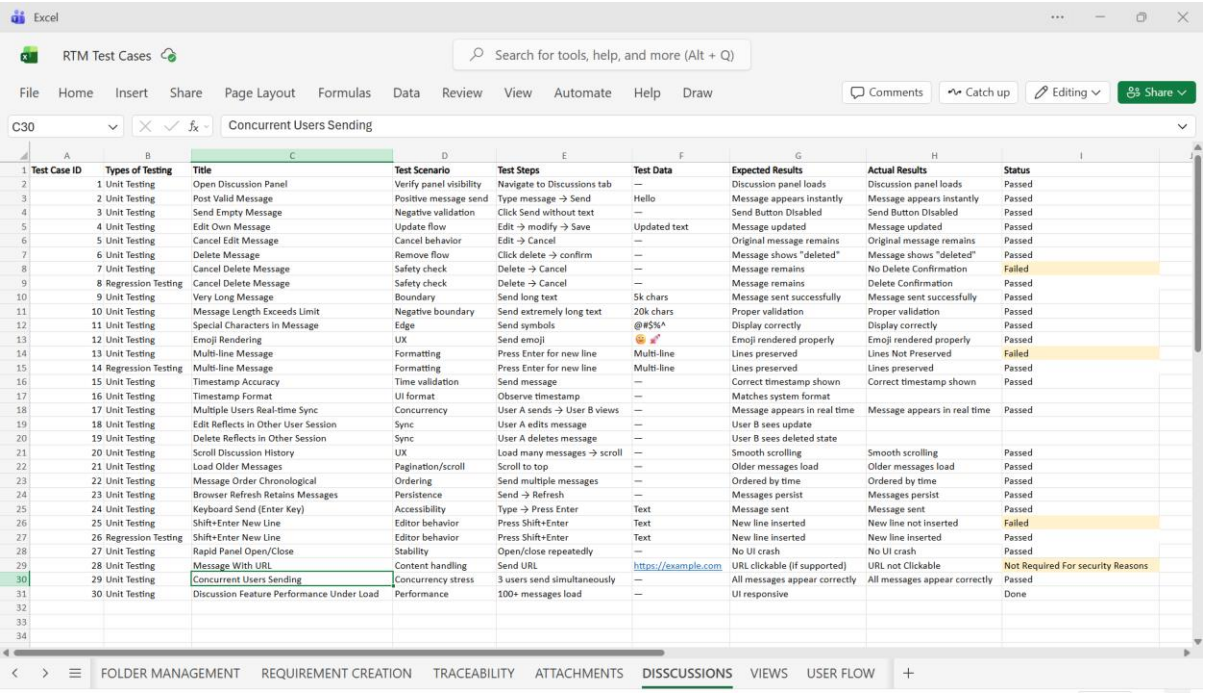
Use Playwright automation for end-to-end testing.





Task 3:

RTM Testing – Did manual testing and recorded and maintain the list of test cases and scenarios.



Test Case ID	Types of Testing	Title	Test Scenario	Test Steps	Test Data	Expected Results	Actual Results	Status
1	Unit Testing	Open Discussion Panel	Verify panel visibility	Navigate to Discussions tab	—	Discussion panel loads	Discussion panel loads	Passed
2	Unit Testing	Post Valid Message	Positive message send	Type message → Send	Hello	Message appears instantly	Message appears instantly	Passed
3	Unit Testing	Send Empty Message	Negative validation	Click Send without text	—	Send Button Disabled	Send Button Disabled	Passed
4	Unit Testing	Edit Own Message	Update flow	Edit → modify → Save	Updated text	Message updated	Message updated	Passed
5	Unit Testing	Cancel Edit Message	Cancel behavior	Edit → Cancel	—	Original message remains	Original message remains	Passed
6	Unit Testing	Delete Message	Remove flow	Click delete → confirm	—	Message shows "deleted"	Message shows "deleted"	Passed
7	Unit Testing	Cancel Delete Message	Safety check	Delete → Cancel	—	Message remains	No Delete Confirmation	Failed
8	Regression Testing	Cancel Delete Message	Safety check	Delete → Cancel	—	Message remains	Delete Confirmation	Passed
9	Unit Testing	Very Long Message	Boundary	Send long text	5k chars	Message sent successfully	Message sent successfully	Passed
10	Unit Testing	Message Length Exceeds Limit	Negative boundary	Send extremely long text	20k chars	Proper validation	Proper validation	Passed
11	Unit Testing	Special Characters in Message	Edge	Send symbols	@#$%^	Display correctly	Display correctly	Passed
12	Unit Testing	Emoji Rendering	UX	Send emoji	😊 💕	Emoji rendered properly	Emoji rendered properly	Passed
13	Unit Testing	Multi-line Message	Formatting	Press Enter for new line	Multi-line	Lines preserved	Lines Not Preserved	Failed
14	Regression Testing	Multi-line Message	Formatting	Press Enter for new line	Multi-line	Lines preserved	Lines preserved	Passed
15	Unit Testing	Timestamp Accuracy	Time validation	Send message	—	Correct timestamp shown	Correct timestamp shown	Passed
16	Unit Testing	Timestamp Format	UI format	Observe timestamp	—	Matches system format		
17	Unit Testing	Multiple Users Real-time Sync	Concurrency	User A sends → User B views	—	Message appears in real time	Message appears in real time	Passed
18	Unit Testing	Edit Reflects in Other User Session	Sync	User A edits message	—	User B sees update		
19	Unit Testing	Delete Reflects in Other Session	Sync	User A deletes message	—	User B sees deleted state		
20	Unit Testing	Scroll Discussion History	UX	Load many messages → scroll	—	Smooth scrolling	Smooth scrolling	Passed
21	Unit Testing	Load Older Messages	Pagination/scroll	Scroll to top	—	Older messages load	Older messages load	Passed
22	Unit Testing	Message Order Chronological	Ordering	Send multiple messages	—	Ordered by time	Ordered by time	Passed
23	Unit Testing	Browser Refresh Retains Messages	Persistence	Send → Refresh	—	Messages persist	Messages persist	Passed
24	Unit Testing	Keyboard Send (Enter Key)	Accessibility	Type → Press Enter	Text	Message sent	Message sent	Passed
25	Unit Testing	Shift+Enter New Line	Editor behavior	Press Shift+Enter	Text	New line inserted	New line not inserted	Failed
26	Regression Testing	Shift+Enter New Line	Editor behavior	Press Shift+Enter	Text	New line inserted	New line inserted	Passed
27	Unit Testing	Rapid Panel Open/Close	Stability	Open/close repeatedly	—	No UI crash	No UI crash	Passed
28	Unit Testing	Message With URL	Content handling	Send URL	https://example.com	URL clickable (if supported)	URL not Clickable	Not Required For security Reasons
29	Unit Testing	Concurrent Users Sending	Concurrency stress	3 users send simultaneously	—	All messages appear correctly	All messages appear correctly	Passed
30	Unit Testing	Discussion Feature Performance Under Load	Performance	100+ messages load	—	UI responsive		Done

RTM Test Cases - https://kaartechit-my.sharepoint.com/:x:/g/personal/slipsa_kaartech_com/IQBwBW9A2-FURo3wvOtGvNCaAT0fpHZEJzgtrRO0EMUrclg?e=EUZ0kc

Task 4:

Created AWS quality code pipeline for Audit Tracker.

Learnings:

- I understood how to design an end-to-end CI/CD pipeline, integrating stages like source, build, test, and deployment.
- Learned how to use AWS CodePipeline to automate the software delivery process and reduce manual intervention.
- Gained knowledge of integrating quality checks and testing into the pipeline to ensure only validated code moves forward.
- Learned how to manage pipeline triggers, ensuring that updates in the code repository automatically initiate the pipeline.
- Got exposure to debugging pipeline failures and analyzing logs to identify and fix issues.
- Improved understanding of how DevOps and QA are interconnected, especially in maintaining code quality through automation.